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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	Fool's Horizon
Selected Domain	Domain 1 — Aiversecity Future-Ready Workforce
Selected SDG Track (1–17)	4 — Quality Education
Project / Solution Name	FuturePath Intelligence
Name (Leader)	Teh Yu Zhan
Email & Phone (Leader)	tehyuzhan123@gmail.com | 0103097848
University Affiliation(s)	Universiti Tunku Abdul Rahman (UTAR), Kampar Campus
Number of Team Members	1

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Malaysian learners and early-career jobseekers face fragmented career information. Skill requirements, training resources and changing technology are difficult to compare, while black-box recommendations may hide weak evidence. Malaysia's Department of Statistics reports 35.2% skills-related underemployment among employed people with tertiary education in Q2 2026 (DOSM, Labour Force Survey, lfs_qtr_sru_sex: https://data.gov.my/data-catalogue/lfs_qtr_sru_sex). This statistic is broad problem context, not a measured FuturePath outcome. FuturePath focuses on a narrower decision problem: helping learners describe their skills, distinguish confirmed gaps from uncertainty, compare possible roles and identify practical learning actions. We intend to assess usefulness through transparent scoring, user-comprehension tasks and usability testing rather than claiming that our prototype has reduced unemployment.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

FuturePath Intelligence is a web-based career decision-support prototype. Users select or edit skills in four states—known present, known absent, unknown and not reported—then compare role requirements using deterministic, explainable matching. The application distinguishes confirmed skill gaps from missing evidence, offers career comparisons and user-directed learning roadmaps, and labels current public occupation data as synthetic. A public Streamlit prototype has been published on Replit: https://futurepath-sdc-2026-public.replit.app. Python, reproducible GitHub releases and a sanitized Replit deployment demonstrate Phase 1 cloud application skills. Planned AI assistance would explain matching results, coach skill gaps and support a conversational roadmap through a RunPod-hosted GLM endpoint, subject to real API verification, consent and usage controls. Do not regard planned GenAI capability as live unless separately tested and deployed. Pending O*NET occupational datasets and OpenAlex trend data remain outside public scoring until review and licensing gates are cleared. FuturePath is decision support, not an automated hiring decision or verified forecast of job demand.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Python 3.12 and Streamlit 1.66.1 provide the public interface on Replit Autoscale. The deterministic domain layer supplies evidence-aware career matching, gap identification, comparison and offline learning roadmaps using three fictional career profiles and synthetic skill requirements. GitHub maintains source history. This solution was built during the SDC Hackathon period, with GenAI, database architecture and cloud deployment workshop learning applied to design and verification. AI adapters were developed and locally evaluated (Ollama), and an official sponsor-provided RunPod GLM-5.3 endpoint was successfully called once in a bounded API test. AI guidance and CV upload remain disabled in the public version; no database of real labour-market records is deployed.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We prioritized a transparent prototype that users can understand and verify. Four skill evidence states—present, absent, unsure, and unreported—prevent a missing answer from becoming a false skill gap. Incomplete evidence can suppress misleading scores. We tested deterministic scoring and user-session isolation, reviewed the workflow in browser sessions, and prepared a public Streamlit deployment with a separate GitHub repository. Our Phase 2 plan is to validate data provenance, conduct usability studies, quantify completion time, and extend AI assistance only with documented safeguards. No employment outcomes or impact improvements have yet been measured.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	SDG Target 4.4: substantially increase youth and adult technical/vocational skills for employment and decent jobs. SDG 8.6 is a secondary intended connection; no measured impact claimed.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	[SOURCE CHECK BEFORE SIGNING] Earlier research notes cite 35.2% skills-related underemployment among employed tertiary-educated people (Malaysia Q2 2026, DOSM); the source and definition must be verified. This contextual statistic is not a measured FuturePath baseline. Phase 2 will directly measure manual task completion time.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Pilot KPI: median minutes to identify one confirmed skill gap and a feasible learning next step; secondary KPI: proportion correctly interpreting unknown versus absent skills. Collect both against a pre-defined comparison task.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Phase 2 prospective TARGET (not observed): ≥20% reduction in median task completion time versus a manual-information baseline in a documented pilot. Record sample size, timing method and confidence limits.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☒ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☐ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☒ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☐ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☒ No — Team leader confirms no pre-existing project work was reused.

If yes, briefly describe what was pre-existing and which team member(s) owned it:

Not applicable — no pre-existing code, design, or research is declared by the team leader.

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☒ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

YES — Python ecosystem including Streamlit, pandas, NumPy, scikit-learn, requests and other locked libraries; licenses must be cross-checked against THIRD_PARTY_NOTICES.md. Synthetic fixtures are created for this project. RunPod GLM must only be listed if actually used.

Name	Type (code/dataset/model)	License

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☒ No — Team leader confirms internship agreement checked and no ownership conflict identified.

If yes, please describe the nature of the agreement and which member it concerns:

Team leader confirms the internship agreement has been reviewed and no IP ownership conflict was identified for this independently developed competition project. [FINAL REVIEW: confirm no employer-owned assets, confidential data or employer equipment were used.]

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

GiaN

Date: 25 October 2026